FFF-F-486B
October 28, 1976
SUPERSEDING
FFF-F-00486A(DSA-DM)
February 15, 1967
FFF-F-486
March 13, 1945 and
MIL-F-36544A
June 10, 1974
(see 6.5)

FEDERAL SPECIFICATION

FLOSS, DENTAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental floss suitable for use in dental operative procedures and in preventive dentistry.

1.2 Classification.

1.2.1 Types and classes. The floss covered by this specification shall be of the following types and shapes as specified (see 6.1):

Type I - Waxed

Shape 1 - Round
Shape 2 - Tape

Type II - Bonded

Shape 1 - Round

Type III - Natural

Shape 1 - Round

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Source: https://assist.dla.mil -- Downloaded: 20260-26T12:36Z
Check the source to verify that this is the current version before use.

Federal Specifications:

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Federal Standards:

Fed. Std. No. 123	- Marking for Shipment, (Civil Agencies)
Fed. Std. No. 191	- Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-D-36242	- Dispenser, Dental Floss, Metal.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:36Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D276 Identification of Fibers in Textiles.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

The United States Pharmacopeial Convention, Inc.:

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

American Pharmaceutical Association:

National Formulary

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Yarn. The yarn shall be natural, white dental floss, made of bright, high tenacity, continuous, multi-filament nylon.

3.1.2 Waxing agent. The wax used for type I floss shall be White Wax, U.S.P., or Yellow Wax, N.F., or highly refined petroleum wax, or combination of the aforementioned waxes. The petroleum wax used shall have a melting point between 135° Fahrenheit (F.) and 180°F., and when heated, the wax shall reduce to a clear, colorless liquid free from foreign particles. The wax content shall be 15 to 30 percent for shape 1 and 20 to 40 percent for shape 2, respectively, when tested as specified in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:36Z
Check the source to verify that this is the current version before use.

3.1.3 Bonding agent. The bonding agent used in type II, shape 1 floss and as an alternate, for type III, shape 1 floss, shall be a colorless nylon polymer suitable for use in the oral cavity.

3.2 Description. The floss shall be suitable for use in dental operative procedures. The floss shall be furnished in the type and shape shown in 1.2 and length shown in 3.3.1 as specified in the procurement document (see 6.1). The floss shall conform to the requirements of table I for number of filaments, denier of ends and yarn, and yarn twist. Type I floss shall be waxed nylon yarn. Type II floss shall be a nylon yarn bonded with a colorless nylon polymer. Type III floss shall be a natural (unwaxed and unbonded) nylon yarn. Shape 1 floss shall have a round cross section. Shape 2 floss shall have a flat tape cross section measuring 1/16 to 1/32 inch in width.

TABLE I. Yarn Characteristics

Classification	Number of filaments per end	Denier of End	Denier of yarn	Yarn twist (t.p.i.)
Type I, shape 1	30-38	30-100	790-850	1-3
Type I, shape 2	30-38	20-140	1330-1470	1 max.
Type II, shape 1	30-38	70-100	730-815	1-3
Type III, shape 1 1/	30-38	70-100	470-530	1-3

1/ When specified (see 6.1), a bonded floss shall be acceptable as an alternate to the natural floss.

3.3 Put up. Floss shall be uniformly wound on a spool so that the floss unwinds from the outer diameter and each turn and layer is free from entanglement. Unless otherwise specified (see 6.1), the floss shall be furnished in one continuous length as shown in table II for the applicable type and shape.

3.3.1 Types I and II. Unless otherwise specified (see 6.1), types I and II floss shall be furnished as refill items suitable for use in a bracket dispenser. The floss shall be uniformly wound on a cylindrical spool fabricated of rigid plastic, 2-3/4 ± 1/16 inches in length and 5/16 ± 1/32 inch in diameter. The floss shall be wound on the core to a maximum diameter of 1-1/8 inches in such a manner that a minimum of 3/16 inch of the core is exposed on each end of the fully wound spool. The floss shall dispense without snarling or entanglement when inserted in a dispenser in accordance with MIL-D-36242 and tested as specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:36Z
Check the source to verify that this is the current version before use.

TABLE II. Length of Floss

Type	Shape	Minimum Average Length in Yards 1/	Minimum Individual Length in Yards
I	1	200	190
I	2	100	95
II	1	200	190
III	1	100	95

1/ Minimum average length of the floss when sampled and examined in accordance with 4.2.1 and 4.3 shall be as shown in the table above.

3.3.2 Type III. Unless otherwise specified (see 6.1), the floss shall be furnished in a plastic or metal dispenser. The dispenser shall contain the spool of floss and shall have a cap or lid for opening and closing. The dispenser shall have an opening through which the floss may be pulled out and a built-in corrosion resistant metal holding and cutting device for securing and cutting the floss. The floss shall dispense without snarling or entanglement when tested as specified in 4.4.5.

3.3.3 Alternate put-up for type I and II. When specified (see 6.1), the type I or II floss shall be furnished in the put-up as specified in 3.3.2 for the type III floss.

3.4 Breaking strength. The minimum average breaking strength shall be as specified in table III for the applicable types and shapes when tested as specified in 4.4.4.

TABLE III. Breaking strength

Type and shape	Breaking strength (pounds)
Types I, II, Shape 1	9
Type I, Shape 2	18
Type III, Shape 1	6

3.5 Workmanship. Workmanship shall be first class throughout. The floss shall be clean and packaged clean. There shall be no objectionable odor or taste present in the floss. The floss shall be free from lumps, knots, kinks, fuzziness, fraying, tangling, or non smooth surfaces, and other imperfections which may detract from its appearance or impair its serviceability.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:36Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of compliance. Certificates of compliance supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to **check test** such items to determine the validity of the certification.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables IV through VII (see 4.2.1, 4.2.2, 4.3 and 4.3.1), and 4.3.1.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table IV. Unit of product for examination purposes shall be unit of issue.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:36Z
Check the source to verify that this is the current version before use.

TABLE IV. Inspection and acceptable quality levels.

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	2.5
Minor	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table V, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be unit of issue.

TABLE V. Tests.

Characteristic	Requirement	Test procedure	Inspection level
Wax content	3.1.2	4.4.3	S-1
Dispensing of floss	3.3.1 & 3.3.2	4.4.5	S-1
Breaking strength	3.4	4.4.4	S-1

4.2.3 Raw materials. In the absence of a certificate of compliance (see 4.1.3), a sufficient quantity of each lot of the nylon, bonding resin (when applicable) and the wax (when applicable) used in the floss shall be selected for chemical analysis.

4.3 Examination. The floss shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table VI.

TABLE VI. Classification of Defects

Categories	Defects
Major:	
101	Nylon not continuous filament nylon.
102	Wax improperly or unevenly applied (Type I only).
103	Core, spool, or reel on which floss is wound not smooth.
104	Type (waxed, bonded, or natural) and shape (round or tape) not as specified.
105	Floss improperly or not firmly wound on core, spool, or reel.
106	Workmanship defects such as kinks, knots, entangling, fuzziness or non-smooth surface, present in floss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification of Defects (cont'd)

Categories	Defects
107	Floss not clean and not packaged clean; dirty spots or stains clearly visible or noticeable.
108	Objectionable odor or taste present in floss.
109	Number of filaments per end not as specified. 1/
110	Denier ends not as specified. 1/
111	Denier of yarn not as specified. 1/
112	Number of twists (turns per inch) not as specified. 1/
Minor:	
201	Wax not white in color.
202	Dispenser (when furnished) not equipped with suitable cutting device.
203	Floss does not have bright finish.
204	Length of floss not as specified.

1/ Examination for floss defects may be accomplished with any suitable measuring device, microscope, twist tester, balance, vibroscope, or instrument of equal accuracy.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

TABLE VII. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.

4.3.1.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-566, PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636 and PPP-B-676 the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the appropriate box specification.

4.4 Test procedures.

4.4.1 Material. Material identification for nylon shall be conducted in accordance with either the American Society for Testing and Materials (ASTM) Method D276 or Method 1530 of Fed. Std. No. 191.

4.4.2 Physical properties of petroleum wax. Petroleum wax shall be tested for melting point. Wax shall be heated and the liquid observed for color and foreign particles.

4.4.3 Wax content. A 10-yard sample of floss shall be weighed. The sample shall then be soaked in a wax solvent such as carbon tetrachloride until all of the wax has been removed, as determined by the sample arriving at an equilibrium of constant weight. After all of the solvent has been dried off the sample, it shall be reweighed and the percent of wax content calculated as follows:

Percent of wax content =

$$\frac{\text{Weight of sample} - \text{Weight of unwaxed sample} \times 100}{\text{Weight of sample}}$$

A minimum of two determinations shall be made and the results averaged.

4.4.4 Breaking strength. Breaking strength shall be determined by one of the following ~~two~~ methods stated below, in 4.4.4.1 and 4.4.4.2.

4.4.4.1 Method 4100.1 of Fed. Std. No. 191 (for Breaking Strength only).

4.4.4.2 Tests shall be performed in a machine as described in 6.4 or equivalent as approved by the contracting officer, having two clamps whose centers shall be in the same plane parallel with the direction of motion of the stressing clamp and so aligned that they will hold the test specimen wholly in one plane. The lower grip shall be installed in the movable stage of the tensile tester. The upper grip shall be secured to the shorter end of the dial gauge. The lower grip shall be brought to the highest position by

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:36Z
Check the source to verify that this is the current version before use.

moving the stage of the tensile tester to its highest position. The dial gauge shall be clamped to the upper grip. The clamped height shall be adjusted in such a way that the closest distance between the centers of the circular sections of the 2 grips is 2 ± 0.2 inches. (This adjusts the clamped length of the specimens to 2 ± 0.2 inches. The zero reading on the dial shall be adjusted with the upper grip suspended freely. The dial gauge shall then be set in the mode for recording maximum tensile stress. The conditioned test specimen shall be clamped securely between the two grips avoiding sag or excessive tension and insuring that the winding of the test specimen around the circular portions of the grips follows the groove. Specimens shall consist of 40 to 50 cm lengths of the test sample. Specimens shall be mounted on a non-reactive surface at the two ends without applying excessive tension. The specimens shall then be exposed to controlled temperature-humidity environment for a period of at least four hours at 70 F. ± 2 F. at 65% ± 5% relative humidity. The clamped ends of the specimen shall be tightly secured to prevent slippage while under tension. Tension shall be applied to the clamped specimen by pulling the movable grip away at the pre-adjusted elongation rate of two inches per minute. The maximum stress (breaking strength) shall be recorded after the specimen ruptures. At least 5 specimens shall be broken and the results averaged.

4.4.5 Dispensing of floss. A minimum of 3 yards of the floss shall be pulled from the container. The floss shall be examined during and after dispensing for any tangles or snarls.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Type I, shapes 1 or 2 or type II, shape 1. Each refill shall be wrapped in tissue or other suitable material and placed into a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.1.2 Type III, shape 1. Thirty six dispensers shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.

5.1.1.2 Intermediate package. Twenty four (24) units of type 1, shapes 1 or 2 or type II, shape 1 shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636. Closure shall be as specified in the appendix of the applicable box specification. Intermediate package is not required for type III, shape 1 floss.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve (12) intermediate packages of type I, shapes 1 or 2 or type II, shape 1 or twenty four (24) unit packages of type III, shape 1 shall be packed in an exterior shipping container designed for a type load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twelve (12) intermediate packages of type I, shapes 1 or 2, or type II, shape 1, or twenty-four (24) unit packages of type III, shape 1 shall be packed in an exterior shipping container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Units shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Each unit, intermediate and shipping container shall be marked in accordance with Fed. Std. No. 123.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:36Z
Check the source to verify that this is the current version before use.

5.3.2 Military activities. Each unit, intermediate and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and shape required (see 1.2.1).
(c) If length other than that specified (see 3.3.1).
(d) Whether floss is to be furnished as refill item or in a dispenser (see 3.3.1, 3.3.2, and 3.3.3).
(e) Whether bonded floss is required as an alternate to natural floss for type III, shape 1 (see 3.2).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Denier. Denier is defined as the weight in grams of 9000 meters of yarn, e.g. a 200 denier nylon yarn is one wherein 9000 meters of yarn weighs 200 grams.

6.3 Floss dispenser for type I floss. Dispenser conforming to the requirements of NSN 6520-00-890-1871 shall apply.

6.4 Breaking strength. Example of machine which may be used for Breaking Strength determination is Chatillon motorized universal tester model LT-CM with Dial Push Pull Gauge DPP-25 and GF-3 grips.

6.5 Supersession data. This specification includes the requirements of MIL-F-36544A dated 10 June 1974.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6520-1355

Civil Agency Coordinating Activities:
VA-DMS
HEW-HSM
GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

FFF-F-486B - Floss, Dental

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:36Z
Check the source to verify that this is the current version before use.